Course Code: SECP2613-01
Lecturer's Name: DR ARYATI BINTI BAKRI
Assignment 2: Data Flow Diagram

Group Name: Group 7

Group members:

Name	**No Matric**
Muhammad Najmi Shahmi Bin Mohd Latpi	A25CS0279
Nasaaie Bin Noriskamar	A25CS0118
Anaqi Harraz Bin Mohd Azad	A25CS0191
Shasya Shafieqah Binti Shaharuddin	A25CS0350
Puteri Anis Annisa Binti Mat Lazim	A25CS0339

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1. Problem Definition

(a) Current process

Based on the scenario, the MediCare Plus Clinic currently manages appointments using the following methods:

· Phone calls
· WhatsApp messages
· Paper appointment books
· Excel files

(b) Current problems

Based on the scenario, the clinic faces at least four distinct problems:

1. Duplicate booking slots occurring due to the manual and disconnected tracking methods.
2. Difficulty tracking patient queue status because there is no centralized or real-time system.
3. Long waiting times for patients as a result of inefficient queue and appointment management.
4. Delayed reporting and inefficient access to appointment information due to reliance on paper and separate Excel files.

(c) Formal problem statement

The MediCare Plus Clinic currently relies on a manual and semi-digital appointment management system using phone calls, WhatsApp, paper books, and Excel files. As patient volume increases, this fragmented approach has led to operational inefficiencies including duplicate bookings, inability to track real-time queue status, prolonged patient waiting times, and delays in generating reports or accessing appointment data. Therefore, there is a need to investigate the feasibility of implementing a Clinic Appointment and Patient Queue Management System to address these operational problems.

2. Objective Requirement and Constraint

Objectives:

1. Prevent duplicated appointment bookings by implementing a centralized schedule system.
2. Make the process of booking, canceling and rescheduling appointments for both patients and staff more simple.
3. Provide real-time visibility of patient queue status so that clinic staff can manage patient flow effectively and patients can be informed of their estimated waiting time.
4. Automatic generates reports that support management in monitoring clinic performance and data.
5. Improve the retention of the patient by providing a smoother and smarter clinic experience

Functional requirements:

1. Allow booking appointments, reschedules or cancellation through the system by both patients and Medicare Plus Clinic staff.
2. Provide real-time notifications to inform the patients about appointment confirmations, delays, cancellations or schedule changes.
3. Automatically block the time slots once they are booked to prevent double-booking.
4. Generate comprehensive reports to help management to analyze patient flow and clinic utilization.
5. Display real-time patients queue status along with estimated waiting time for both doctors and patients.
6. Allow patients to view available time slots and book appointments online.

Non-functional requirements:

1. Provide high availability to ensure appointment booking can be done at night and when the clinic is closed.
2. Deliver fast response times when performing operations like booking, updating appointments and queue management.
3. Include automated backup and recovery features to prevent data loss.
4. Keeps all patients' information private and only accessible staff can access it.

Constraints:

1. System design must be simple and user-friendly so that clinic staff and elderly patients can operate without requiring technical training.
2. Support multiple user accessing at the same time without causing delay or system errors.
3. Ensure all data is securely stored and protected from unauthorized access.
4. Must be able to run on standard web browsers without requiring the purchase of new hardware.

3. Feasibility Study

- Technical Feasibility

  To make a Clinic Appointment and Patient Queue Management System it is a need to consider the hardware requirements with the budget of RM 15000 as mentioned in the CBA . In order to run the system smoothly, we need a stable Local Area Network and high-speed Wi-Fi as we will use a cloud-based queue system.Besides, the clinic need to have basic hardware like tablets for queue registration and a central PC for the administrator so they can change from record the registration appointment data manually using paper books and Excel files to an automated databases which is well within the budget. Plus, from here the duplicate files problem will be solved. For existing patient data in Excel , the technical team will need to focus on ETL processes which extract , transform and load to transfer the existing data to the new system database without data loss.For system maintenance and security updates , the clinic can consider a third party vendor as it is possible considering the budget of maintenance per year. Overall, from the technical aspect it is feasible for the system to be done as the clinic provides a sufficient amount of budget in hardware and software and the technical requirements for the queue management system can be reached and low-risk.

- Operational Feasibility

  This system will replace all the manual processes and provide real-time visibility for both staff and patients.Hence, the staff and the patients acceptance towards the new system needs to be considered. They might find the system interface is challenging to use initially.Therefore, the clinic can consider implementing a structured training program for staff in order for them to get used to this digital system.For patient side,as the system main focus is to design a system that can reduce waiting process it will improves the patient experience.The patients can gain a convenience self services.Nevetherless,we need to elderly patient as they will face difficulty in adapting to this new technology systems so the manual systems can be continue.

  This system will successfully solve the issues of delay reporting by providing management with instant access to patient data and clinic performances which align with the clinic goals.For the start, it will be partially feasible because although this systems can achieve all the requirements needed from the clinic, it will take time for people to adapt with the new technology.However , as the time goes by, it can change to feasible completely as they will get use to the new system.

- Economic Feasibility

With the estimated cost provided by the clinic, it shows that it is possible to cover all the purchases of hardware,software and all the new equipment needed to develop the system.
The Cost Benefit Analysis (CBA) was conducted using a 3-year Present Value projection with an 8% discount.When applying the mandatory sensitive factor, it indicates the result of significant wide gap between expenditure and return.On top of that,the resulting result for profitability index is 0.16 which show high financial risk as it shows that the combination of cost for hardware , software , maintenance and support are far outweighs the projected interest and service benefits within the 3 year window.To sum up , it is not feasible because low PI showing that it is not good investment.

- Overall Feasibility decision

While the economic feasibility with a PI of 0.16 suggests a very high financial loss making it too risky for the investment at the moment but the project remains technically and operationally sound. This economic point of view is based purely on a 3 year monetary return.However, from the strategic perspective,it is still a good project to be done as it solves all the critical issues faced by the clinic such as duplicate files , long waiting time and delayed reporting.Therefore,the project is considered as partially feasible if the clinic prioritizes on service quality over immediate short-term profit.

4. Economic Feasibilty: Cost-Benefit Analysis (CBA)

Cost	Year 0	Year 1	Year 2	Year 3
Development Cost				
- Hardware	16500			
- Software	22000			
Total	38500			
Production Cost				
- Maintenance		5500	5775	6063.75
- Support		11000	11550	12127.5
Total		16500	17325	18191.25
Annual Prod Costs (Present Value)		15277.78	14853.40	14440.80
Accumulated Costs		53777.78	68631.18	83071.98

Benefits	Year 0	Year 1	Year 2	Year 3
- Interest investment		10800	11232	11681.28
- Consultation services		18000	18720	19468.80
Total		28800	29952	31150.08
(Present Value)		26666.67	25679.01	24727.94
Accumulated Benefits (Present Value)		26666.67	52345.68	77073.62
Gain / Loss		(27111.11)	(16285.50)	(5998.36)
Profitability Index	0.16			

Justification : Based on the value of the Profitability Index which is 0.16.This project should not proceed as the value is less than one.The value considered economically unprofitable

5. Conclusion

Based on the feasibility analysis of The MediCare Plus Clinic Appointment and Patient Queue Management System show that the project has its own advantages and weak points. This project can still proceed because the necessary hardware, software, and network infrastructure can continue with the available budget. The use of cloud-based systems with proper data migration methods like ETL can solve the current serious issues which are duplicate bookings and bad patient monitoring.

From a project planning perspective, these systems can also be considered feasible. By shifting to this system, the clinic can provide a more structured and reliable solution. Other than that, the clinic may also achieve a better centralized data management and real-time queue tracking systems, which can lead to efficient operation.

However, Cost-Benefit Analysis results in a Profitability Index (PI) of 0.16 over the three years, which is below the standard benchmark of 1.0. This situation shows that it is not good enough financially, especially in the short term. Despite this, the system has its own strength which can improve patient satisfaction, better service quality and more efficient clinic management.

In conclusion, although the system is not financially viable in terms of profit, it is a good decision to proceed with this project due to its long-term benefits, quality and efficiency.

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